



NONLINEAR MODEL

UPA862TD

BJT NONLINEAR MODEL PARAMETERS⁽¹⁾

Parameters	Q1 NE685	Q2 NE851	Parameters	Q1 NE685	Q2 NE851
IS	7.0e-16	137e-18	MJC	0.34	0.14
BF	109	166	XCJC	0.5	0.5
NF	1	0.9871	CJS	0	0
VAF	15	20.4	VJS	0.75	0.75
IKF	0.19	50	MJS	0	0
ISE	7.90e-13	80.4e-15	FC	0.1	0.55
NE	2.19	2.4	TF	2.0e-12	18e-12
BR	1	28.7	XTF	6	0.1
NR	1.08	0.9889	VTF	3	2
VAR	12.4	2.7	ITF	0.005	0.03
IKR	infinity	0.021	PTF	0	0
ISC	0	532e-18	TR	1.0e-9	1.0e-9
NC	2	1.28	EG	1.11	1.11
RE	1.3	0.45	XTB	0	0
RB	5	4	XTI	3	3
RBM	3	1	KF (2)	0	0
IRB	0.005	0	AF (2)	1	1
RC	10	1.7			
CJE	0.4e-12	2.4e-12			
VJE	0.81	0.87			
MJE	0.5	0.34			
CJC	0.18e-12	0.65e-12			
VJC	0.75	0.52			

(1) Gummel-Poon Model

(2) AF and KF are 1/f noise parameters and are bias dependant.
The appropriate values for the 1/f noise parameters (AF and KF) shall be chosen from the table below, according to the desired current range.

Q1		Ic = 5 mA	Ic =10 mA	Ic = 15 mA
	KF	54.38e-12	997.6e-12	500.2e-12
	AF	2.071	2.375	2.288

Q2		Ic = 5 mA	Ic =10 mA	Ic = 15 mA
	KF	4.547e-15	855e-12	1.73e-9
	AF	1.4	2.551	2.626

MODEL RANGE

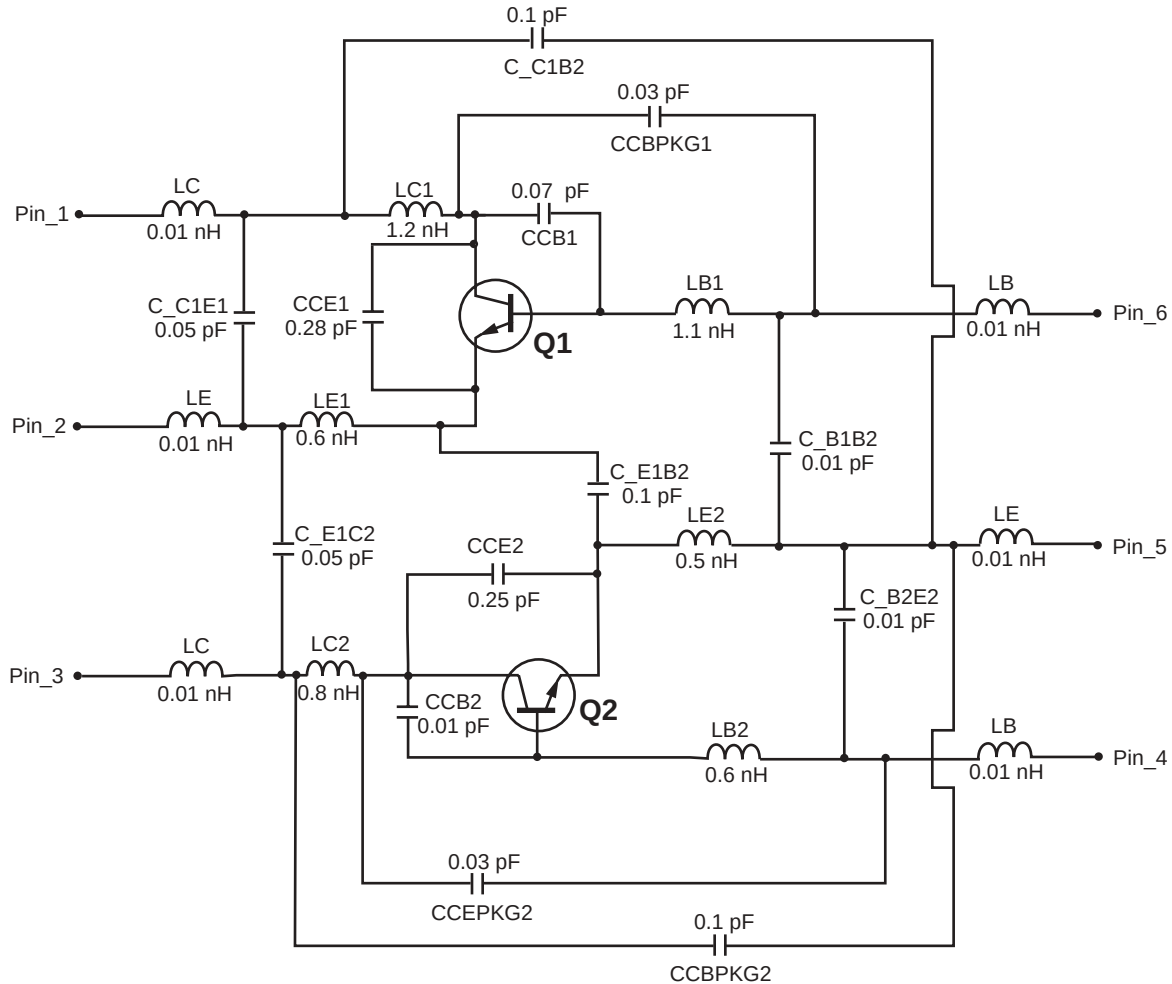
Frequency: 0.1 to 3.0 GHz

Bias: VCE =0.5 V to 2.5 V, Ic = 1 mA to 20 mA

Date: 08/03

For a better understanding on AF and KF parameters, please refer to AN1026.

SCHEMATIC



MODEL RANGE

Frequency: 0.1 to 3.0 GHz

Bias: $V_{CE} = 0.5 \text{ V to } 2.5 \text{ V}$, $I_c = 1 \text{ mA to } 20 \text{ mA}$

Date: 08/03

Life Support Applications

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